MSC IN ARTIFICIAL INTELLIGENCE · APPLIED MACHINE LEARNING

Predicting Geographic Provenance from Bird-Observation Covariates

A stacked heterogeneous ensemble over 1.55 million Sri Lankan biodiversity records, with a Bayes-error analysis that redefined the task and a three-method explainability audit.

AUTHOR

Hansika Gunasekara

STUDENT ID

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DATASET

`DilushaChandrasiri/SriLanka-Bird-Diversity-Dataset (Hugging Face)`

SOURCE CODE

`github.com/HansikaGunasekara/srilanka-bird-diversity-ensemble`

LIVE APPLICATION

`srilanka-bird-diversity-ensemble.streamlit.app`

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Abstract

This report applies a stacked heterogeneous ensemble to 1,552,048 bird-observation records from Sri Lanka, each joined to sixteen remotely-sensed environmental covariates. The assignment was originally framed as species classification across 429 classes. A Bayes-error analysis conducted before any modelling showed that framing to be *provably* unsolvable: because every environmental covariate is a deterministic function of the record's location and date, the 1.55 million rows collapse to only 63,392 distinct feature vectors, and roughly twenty-four rows share a byte-identical input while carrying different species labels. Top-1 species accuracy is therefore capped at 5.54% for any classifier that will ever be built on this feature set. The task was accordingly re-specified as predicting `stateProvince` — one of twenty-five administrative districts — whose measured ceiling on the same features is 100%. A six-member ensemble combining bagging, histogram-based boosting and a linear model, fused by accuracy-weighted soft voting and by a logistic meta-learner trained on three-fold out-of-fold probabilities, reaches **0.9995** accuracy under the conventional stratified split. Because province correlates strongly with latitude and longitude, three independent guards against spatial memorisation were run; the strictest of them — unseen test sites *and* coordinates withheld — yields **0.9032**, below the assignment's 95% threshold, and is reported rather than suppressed. A three-method explainability audit (TreeSHAP, permutation importance and partial dependence) shows the model's learned structure aligns with Sri Lankan physical geography, and surfaces a quantified disagreement between importance methods that is itself the report's most interesting interpretive finding.

1 Problem Definition and Dataset Collection

1.1 The problem and why it matters

Sri Lanka is one of the world's thirty-six biodiversity hotspots, and bird communities are a standard indicator taxon for habitat condition: they are conspicuous, well-recorded by volunteers, and respond quickly to land-use change. The island's biodiversity record is dominated by citizen-science checklists, which are abundant but spatially uneven.

The applied question this project addresses is: **given only the environmental and temporal signature of an observation — its climate, terrain, vegetation, aerosol chemistry and date — can a model recover *where in Sri Lanka* that observation came from?** A model that can do so has three concrete uses. First, *record validation*: national biodiversity databases carry a persistent rate of transcription and georeferencing errors, and a classifier that disagrees confidently with a record's stated district flags it for human review. Second, *provenance inference* for historical or digitised records whose locality is vague ("Central Province, 1974"). Third, and most interesting scientifically, the model is a measuring instrument for **environmental distinctiveness**: if a district's ecological fingerprint can be recognised without its coordinates, that district is environmentally distinct, and the magnitude of the accuracy drop when coordinates are removed quantifies how much of its identity is geographic accident versus genuine habitat character.

1.2 Data source and how it was collected

The dataset is `DilushaChandrasiri/SriLanka-Bird-Diversity-Dataset`, published openly on the Hugging Face Hub. It was obtained programmatically with `huggingface_hub.snapshot_download` (see `src/download.py`), which pins a dataset revision and makes the acquisition step reproducible rather than a manual download.

The file is a **derived, pre-joined product** rather than a raw export, and understanding that provenance turned out to be decisive for this project. Its two layers are:

- **Occurrence layer.** Darwin-Core-style species occurrence records (`verbatimScientificName`, `individualCount`, `decimalLatitude`, `decimalLongitude`, `eventDate`, `stateProvince`), consistent with a GBIF-mediated aggregation of citizen-science checklists.
- **Environmental layer.** Sixteen covariates attached to each record by spatial join from satellite and reanalysis products — VIIRS night-time lights, MODIS NDVI and land cover, a digital elevation model, MERRA-2 aerosol species, and gridded climate normals.

The critical consequence: the environmental layer is a deterministic lookup keyed on (latitude, longitude). Two records from the same place on the same day necessarily carry identical environmental values. This is not a defect of the dataset — it is how such products are built — but it governs what can and cannot be predicted from it, as §1.5 establishes.

1.3 Size, features and target variable

1,552,048	24	19,384	2014–2024	25
occurrence records	model features	distinct sites	temporal span	target classes

Group	Columns	Description and role
Geographic (2)	`decimalLatitude, decimalLongitude`	WGS-84 coordinates. Deliberately isolated so they can be withheld as an ablation (§3.5).
Climate (5)	`temp_mean, rainfall, wind_mean, humid_mean, shortwave_radiation`	Gridded climate normals at the observation point.
Terrain & vegetation (4)	`elevation_meters, NDVI, LandCover_Class, lka_general_2020`	DEM elevation, greenness index, categorical land-cover code, modelled population density.
Aerosol chemistry (6)	`Carbon_Mass, Dust_Mass, SO2_Mass, Sulfate_Mass, Sea_Salt_Mass, Total_Aerosol_Extinction`	MERRA-2 column masses. `Sea_Salt_Mass` proves an informative marine-proximity proxy (§4.4).
Light (1)	`avg_rad`	Night-time radiance — an effective urbanisation proxy.
Temporal (5, engineered)	`year, month, doy, sin_doy, cos_doy`	Decomposed from `eventDate`; see §1.4.
Observation metadata (1)	`individualCount`	Retained as a legitimate column; contributes almost nothing (§4.3).
Target: `stateProvince` — 25 administrative districts. Severely imbalanced: Colombo 18.64% of records, Vavuniya 0.12% — a **161×** ratio. This imbalance dictates the choice of evaluation metrics (§3.4) and is the basis of the fairness analysis in §5.3.		

1.4 Preprocessing

Six deliberate steps take the raw file to the matrix the models see. Each is implemented in `src/prep.py`.

Step	Operation	Justification
Date decomposition	Parse `eventDate` → `year`, `month`, `doy`	A tree cannot split usefully on a raw timestamp. It can split on season and on long-run trend.
Cyclical encoding	`sin_doy = sin(2π·doy/365.25)`, likewise cosine	Under a linear encoding, 31 December (day 365) and 1 January (day 1) sit at opposite extremes though they are one day apart. Projecting onto the unit circle restores adjacency, so a split near the year boundary is expressible.
Memory downcast	float64 → float32; cache as Parquet	Halves the working set, letting 1.55 M × 24 fit comfortably in 8 GB. Reload drops from ~90 s (CSV parse) to ~2 s.
Leakage audit	Drop the `index` column after testing it	A monotonic row identifier can encode the label if the file was written grouped by class. Tested explicitly: adjacent rows share a species 1.04% of the time, versus 1/429 = 0.23% under independence — a weak ordering effect, not a usable leak. Dropped regardless.
Two feature sets	`full` (24) and `envonly` (22)	A built-in ablation. `envonly` removes the coordinates so the accuracy claim cannot rest on a point-in-polygon lookup.
Targeted scaling	`StandardScaler` inside the logistic-regression pipeline only	Tree ensembles are invariant to monotone rescaling; logistic regression is not. Scaling is applied only where it changes the fit, and fitted inside the pipeline so the scaler never sees test rows.

No imputation was required: the delivered file has no missing values in the modelled columns. No categorical encoding was required either — `LandCover_Class` arrives as an integer code, which tree learners handle natively as an ordinal split variable.

1.5 The analysis that changed the target variable

> **Species classification on this feature set is capped at 5.54%, for any model**
>
> Before fitting anything, an irreducible-error (Bayes) analysis was run (`src/ceiling.py`). Because every environmental covariate is a deterministic function of (latitude, longitude, date), the feature matrix has far less diversity than the row count suggests:
>
> - 1,552,048 rows collapse to **63,392 distinct (lat, lon, date) keys** — about **24 rows per key**.
> - Rows sharing a key are *byte-identical in every feature* while carrying different species.
> - The best any classifier can do on a group of identical inputs is predict that group's most frequent label. Summing those best cases over all groups gives the **Bayes-optimal accuracy**.
>
> Measured ceilings: **0.0554** for species on environment+geography+time; 0.3102 even if `individualCount` is added; 0.9602 if the problem is relaxed to the fifty commonest species. For `stateProvince` the ceiling is **1.0000**. The assignment's 95% requirement is therefore unreachable under species classification by a factor of seventeen, and perfectly reachable under province classification. **The target was changed on this evidence, not for convenience.**

This is a statement about *this feature set*, not about the difficulty of bird identification. Species recognition is highly solvable from images or audio; it is simply not recoverable from the environmental covariates of a shared checklist, because two birders standing in the same place on the same morning record different species from identical environmental conditions. The re-specified task — predicting provenance — is not a weaker substitute but a different and genuinely answerable question, with the practical applications set out in §1.1.

1.6 Ethical data use

The dataset was audited column-by-column for personal and sensitive content before use.

- **No personal data is present.** The twenty-four delivered columns contain no observer name, identifier, email, institution or licence field. Darwin Core's `recordedBy` and `recordedByID` — the fields that would carry contributor identity — are absent from this derivative. No individual is identifiable from any record, so no consent question arises and GDPR-style obligations are not engaged.
- **The data is openly licensed and openly published** on the Hugging Face Hub, and is used here for non-commercial academic work consistent with that publication.
- **One genuine sensitivity exists and is handled.** Precise coordinates for threatened species are a recognised poaching and disturbance risk, which is why GBIF supports coordinate generalisation for sensitive taxa. This project does not publish any raw record. The deployed application ships a 600-row anonymised sample (`artifacts/samples.csv`) used solely to populate example inputs, and the sample carries no species name attached to a precise locality.
- **Attribution.** The underlying observations are the work of volunteer recorders. The dataset, its Hugging Face author and the citizen-science origin are credited in the application's *About the data* tab and in this report.

2 Selection of a New Machine Learning Algorithm

2.1 The algorithm selected

The method applied is **stacked generalisation** (Wolpert, 1992) — also called *stacking* — over a deliberately heterogeneous pool of six base learners, with **histogram-based gradient boosting** (LightGBM, and its scikit-learn and XGBoost analogues) as the dominant members of that pool.

Neither component is a standard taught model, and the combination is the submitted contribution:

- **Stacking** replaces a fixed combination rule with a *learned* one. A second-level model — here multinomial logistic regression — is trained to map the concatenated class-probability vectors of the base learners onto the true label. With six learners and twenty-five classes the meta-learner's input is a 150-dimensional vector per row. Crucially, those inputs must be **out-of-fold** predictions: each base learner's prediction for a training row is produced by a copy of that learner which never saw the row. Without this the meta-learner observes the base models' memorised training fit and learns to trust an optimism that will not survive deployment.
- **Histogram-based gradient boosting** bins each continuous feature into at most 255 discrete buckets before any tree is grown, turning split-finding from a sort over n values into a scan over 255 bins and reducing per-split cost from O($n \log n$) to O(bins). LightGBM adds two further mechanisms absent from a classical boosted tree: **GOSS** (Gradient-based One-Side Sampling), which keeps all large-gradient instances and randomly subsamples the small-gradient ones, and **EFB** (Exclusive Feature Bundling), which merges mutually-exclusive sparse features into a single bundled column. It also grows **leaf-wise** rather than level-wise, splitting whichever leaf offers the greatest loss reduction, which produces deeper, more asymmetric trees at equal leaf count.

2.2 Why this algorithm was selected

1. **The error structure of the problem calls for heterogeneity.** The task is near-separable in the geographic features but genuinely hard in the environment-only ablation. Bagged forests reduce variance; boosted ensembles reduce bias; a linear model contributes a globally smooth decision surface that trees cannot express. Their errors are decorrelated, which is the precondition under which ensembling helps at all. The measured spread across base learners — 0.5448 to 0.9008 on the hardest protocol — confirms the pool is genuinely diverse rather than six views of one model.
2. **A learned combiner outperforms a fixed one where base quality varies by class.** Soft voting applies one weight per model across all twenty-five provinces. Stacking can learn that LightGBM should be trusted for highland districts while the forest is better for coastal ones. The effect is visible where it matters most: on the hardest protocol stacking improves balanced accuracy over soft voting (0.8637 vs 0.8576) and macro-F1 (0.8740 vs 0.8683) even where plain accuracy is marginally lower — that is, it helps precisely on the small classes that plain accuracy ignores.
3. **Scale.** Histogram binning is what makes 1.55 million rows × 24 features tractable on a laptop. LightGBM fits the full six-way problem in 19.6 seconds where the exact-split alternative is an order of magnitude slower.
4. **It satisfies the assignment's constraints exactly:** it is not a deep-learning model, it is not an image-processing application, and it is not a model covered in lectures.

2.3 How it differs from the standard models

Standard model	What it does	How the submitted method differs
Decision tree	One greedy recursive partition; each split chosen to maximise local purity.	A single tree is a high-variance estimator that commits irrevocably to its first split. The submitted method fits *hundreds* of trees under two different regimes — independent and randomised (bagging) versus sequential on the residual gradient (boosting) — and then learns, at a second level, how much to trust each regime per class. A tree gives one partition; this gives a learned consensus over many.
Logistic regression	One linear decision boundary per class in the raw feature space.	Logistic regression appears here only as the *meta-learner*, operating not on the 24 raw features but on the 150-dimensional simplex of base-model probabilities. Its job is inverted: it does not classify observations, it arbitrates between classifiers. The difference is measurable — as a base learner on the `envonly` set it scores 0.5448, while the same algorithm as a meta-learner contributes to a 0.9028 result.
k-Nearest Neighbours	Non-parametric; classifies by majority vote among the *k* closest training points under a fixed metric.	k-NN has no training phase, stores the entire dataset, and degrades badly with 24 dimensions of heterogeneous scale. The submitted method compresses 1.55 M rows into a fixed set of split thresholds, predicts in ~55 ms, and learns a feature-adaptive partition rather than relying on one global distance metric. On this data a metric-based method would also be dominated by the two coordinate columns, which is the exact failure mode the project is designed to test against.
Random Forest (included as a base learner, not the contribution)	Bagged trees, each on a bootstrap sample with a random feature subset at each split; averaged.	A forest averages with *equal* weight and all its members share one inductive bias. Stacking admits structurally different learners and learns unequal, class-conditional weights for them. The forest is one of six voices here, not the method.

3 Model Training and Evaluation

3.1 Sampling

All reported runs train on a **300,000-row stratified sample** drawn with a fixed seed from the 1.55 million available. The constraint is memory: the study-grade forests unpickle to roughly 460 MB and peak well above that during fitting. A sample-size sweep showed accuracy flat to three decimal places between 300k and the full corpus on the random protocol, so the sample costs nothing measurable on the headline number; §5.1 notes where it may still cost something on the rarest classes.

3.2 Train / validation / test split

The protocol is a three-way separation, but the validation stage is implemented as cross-validation rather than as a single fixed block — a choice worth stating explicitly because it affects how the numbers should be read.

Partition	Size	Mechanism	Purpose
Test	75,000 (25%)	`train_test_split(test_size=0.25, stratify=y)`	Held out before any fitting; touched exactly once, to produce every number in §3.5. No model, hyperparameter or threshold was chosen using it.
Train	225,000 (75%)	the complement	Fits the six base learners.
Validation (stacking)	3 × 75,000	`StratifiedKFold(n_splits=3)` inside the training set	Generates out-of-fold probabilities. Every training row receives a prediction from a model that did not see it; these are the meta-learner's inputs. This *is* the validation set for the stacking stage, used three times over rather than once.
Validation (early stopping)	12% of train	`validation_fraction=0.12` inside `HistGradientBoostingClassifier`	An internal held-out slice, carved from the training set only, that decides when to stop boosting (`n_iter_no_change=25`).

Stratification preserves the 161× class imbalance in every partition, so the rarest province (Vavuniya) retains 84 test rows — enough to estimate recall, though with wide uncertainty, as §5.3 quantifies.

3.3 Hyperparameter choices

Hyperparameters were set by reasoned construction and targeted diagnosis rather than by an automated search; §5.1 records this as the project's clearest methodological limitation. The values and their justifications:

Model	Settings	Reasoning
RandomForest **ExtraTrees**	`n_estimators=100, min_samples_leaf=8, max_features="sqrt"`	$\sqrt{p} \approx 5$ of 24 features per split is the standard classification default and maximises inter-tree decorrelation. A leaf floor of 8 caps depth on 225k rows without materially costing accuracy. The deployment build uses `n_estimators=40, min_samples_leaf=50` (§6.2).
HistGradientBoosting	`max_iter=300, learning_rate=0.10, max_leaf_nodes=31, l2_regularization=10.0, early_stopping=True, n_iter_no_change=25`	31 leaves is the conventional depth-5-equivalent for histogram boosters. The learning rate of 0.10 with early stopping lets the iteration count be chosen by data rather than asserted. **`l2_regularization=10.0` is load-bearing, not cosmetic** — see the diagnosis below.
XGBoost	`n_estimators=400, learning_rate=0.15, max_depth=9, subsample=0.9, colsample_bytree=0.8, tree_method="hist"`	Depth-9 level-wise trees are the complexity analogue of LightGBM's 63 leaf-wise leaves. Row and column subsampling supply the stochastic regularisation that depth alone does not.
LightGBM	`n_estimators=300, learning_rate=0.10, num_leaves=63, reg_lambda=10.0, min_sum_hessian_in_leaf=20.0, colsample_bytree=0.8`	63 leaves exploits leaf-wise growth without unbounded depth. **`reg_lambda` and `min_sum_hessian_in_leaf` are jointly required for this model to converge at all.**
LogisticRegression (base)	`C=1.0, max_iter=600,` with `StandardScaler`	Default regularisation; the iteration cap raised until the multinomial solver converged on 24 scaled features.
LogisticRegression (meta)	`C=10.0, max_iter=1200`	Weaker regularisation deliberately: the meta-learner's 150 inputs are already strongly constrained (six probability simplices), so the risk is underfitting the arbitration rule rather than overfitting it.

A diagnosed failure: divergence, not underfitting

In the first complete run LightGBM scored **0.2541** and HistGradientBoosting **0.8977**, against 0.9984 for a plain random forest. The intuitive reading — underfitting, so add capacity — is wrong, and acting on it makes the problem worse.

An `n_estimators` sweep located the true behaviour: LightGBM's *training* accuracy peaked at **0.9586 at 10 trees** and then *collapsed to 0.3468 by 50 trees*. A model whose training accuracy falls as capacity rises is not underfitting; it is diverging. Mean maximum predicted probability was 0.999999999997.

Mechanism. Province is near-separable in these features, so the softmax saturates within roughly ten boosting rounds. For multiclass log-loss the Hessian of an observation is $p(1-p)$, which vanishes as $p \to 1$. Each leaf value is computed as $-g/h$; as the denominator collapses toward machine epsilon the leaf values explode, and subsequent rounds fit the resulting numerical noise.

Fix. Bound the denominator. `l2_regularization=10.0` adds λ to h so the ratio stays finite, and `min_sum_hessian_in_leaf=20.0` refuses to create leaves whose Hessian mass has already collapsed. Result: **0.2541 → 0.9990** (LightGBM) and **0.8977 → 0.9994** (HistGradientBoosting). These two hyperparameters are worth more than every other tuning decision in the project combined.

3.4 Performance metrics and why these ones

Metric	Why it is reported
Accuracy	The assignment's stated criterion. On its own it is misleading here: predicting only the six largest provinces would already score above 0.60.
Balanced accuracy	Mean per-class recall. Weights Vavuniya (0.12% of rows) equally with Colombo (18.64%), so a model that abandons rare districts cannot hide behind the headline figure.
Macro F1	Unweighted mean per-class F1. Unlike balanced accuracy it also penalises *over*-prediction of a rare class, catching the opposite failure mode.
Weighted F1	Support-weighted; reported alongside macro-F1 so the gap between them measures exactly how much of the performance is carried by the large classes.
Log loss	A proper scoring rule on the probabilities themselves, not on the arg-max. It is the metric that exposes the calibration problem discussed in §5.1 — and the metric on which the diverging boosters failed most visibly.
Confusion matrix + per-class report	Renders *which* districts are mistaken for which. Confusions here are overwhelmingly between geographically adjacent districts, which is the expected and interpretable error structure.
Not reported: ROC-AUC. For 25 classes AUC requires a one-vs-rest or pairwise averaging scheme whose aggregate is dominated by the easy majority classes and is less interpretable here than per-class recall. Log loss serves the same purpose — assessing the probabilities rather than the decisions — without that averaging artefact.	

3.5 Results

Headline: all learners, stratified random split, `full` feature set

Model	Accuracy	Balanced acc.	Macro F1	Weighted F1	Log loss	Fit (s)
RandomForest	0.9984	0.9899	0.9935	0.9984	0.0411	4.9
ExtraTrees	0.9933	0.9767	0.9834	0.9932	0.0991	3.1
HistGradientBoosting	0.9994	0.9972	0.9975	0.9994	**0.0020**	157.0
XGBoost	**0.9995**	**0.9978**	**0.9980**	0.9995	**0.0019**	40.4
LightGBM	0.9990	0.9947	0.9959	0.9990	0.0055	19.6
LogisticRegression	0.9661	0.9192	0.9272	0.9659	0.1210	19.3
SoftVote (ensemble)	**0.9995**	0.9973	0.9978	0.9995	0.0366	–
Stacking (ensemble)	**0.9995**	**0.9975**	**0.9980**	**0.9995**	0.0029	268.7

Two observations. First, the ensemble does not beat its best single member on plain accuracy — XGBoost alone already reaches 0.9995, and on a near-saturated task there is no headroom left to recover. The ensemble's value here is *robustness*, which the harder protocols expose. Second, soft voting's log loss (0.0366) is an order of magnitude worse than stacking's (0.0029) at identical accuracy: averaging six probability vectors, two of which are poorly calibrated, produces a blurred consensus, whereas the meta-learner learns to discount them.

The three guards: is 0.9995 real, or memorised geography?

Province correlates strongly with latitude and longitude, so a sceptical reader should ask whether the model has learned ecology or is performing a point-in-polygon lookup. Three independent guards test this, each removing a different crutch.

Feature set	Protocol	What the guard removes	Accuracy	Balanced acc.	Macro F1
`full`	random	Nothing — the conventional baseline	**0.9995**	0.9975	0.9980
`full`	**grouped**	Every test *site* is unseen in training (`GroupShuffleSplit` on rounded lat/lon)	**0.9925**	0.9910	0.9882
`envonly`	random	Latitude and longitude dropped entirely	**0.9971**	0.9932	0.9942
`envonly`	**grouped**	**Both at once** — unseen sites *and* no coordinates	**0.9032**	**0.8576**	**0.8683**

> **Reading the gap honestly**
>
> Each guard alone costs little: withholding seen sites costs 0.7 percentage points, removing coordinates costs 0.2. Applying both costs **9.6 points**, and **0.9032 falls below the assignment's 95% threshold**. This is reported rather than suppressed because the gap is the project's most informative single number. It says that a substantial part of the headline accuracy comes from recognising *a specific site's* environmental fingerprint — the exact NDVI, elevation and aerosol signature of a place the model has seen before — and not purely from generalisable habitat character. The honest claim this work supports is therefore: **0.9995 for validating records from already-surveyed sites, and roughly 0.90 for a genuinely new location with no coordinates**. Those are different claims with different uses, and conflating them would misrepresent the model.

The `envonly` + grouped condition is also where the ensemble finally earns its keep: the best single model reaches 0.9008 (LightGBM) while soft voting reaches 0.9032 and stacking attains the best balanced accuracy (0.8637) and macro-F1 (0.8740). When the problem stops being saturated, combining decorrelated learners starts to pay.

Per-class behaviour

Under the random protocol per-class recall remains high across the full imbalance range: Colombo (14,038 test rows) attains 1.000, and the rarest district, Vavuniya (84 test rows), attains 0.988. The confusion matrix is near-diagonal, and such off-diagonal mass as exists sits between geographically adjacent districts — Colombo/Gampaha, Galle/Matara — which is the interpretable and expected error structure for a provenance model. The caveat on these rare-class estimates is quantified in §5.3.

4 Explainability and Interpretation

Three explainability methods were applied (`src/explain.py`), chosen so that they *cross-check* rather than merely corroborate one another. The explained model is **LightGBM**: TreeSHAP is exact and tractable on it, it scores 0.9990 alone — within 0.0005 of the full ensemble — and since the meta-learner is a linear reweighting of base probabilities, the base model's attributions are the substantive explanation of the ensemble's behaviour. All figures are computed on 6,000 held-out test rows the model never saw.

Method	Type	What it answers
TreeSHAP	Local → global, exact	Shapley values from cooperative game theory: the unique attribution satisfying local accuracy, missingness and consistency. Gives a signed contribution per feature *per prediction*.
Permutation importance	Global, model-agnostic	The drop in held-out accuracy when one column is shuffled. Measures what the *fitted* model actually relies on.
Partial dependence	Global, marginal	The average predicted probability as one feature is swept across its range — the *direction and shape* of an effect, not just its magnitude.

4.1 Which features are most influential

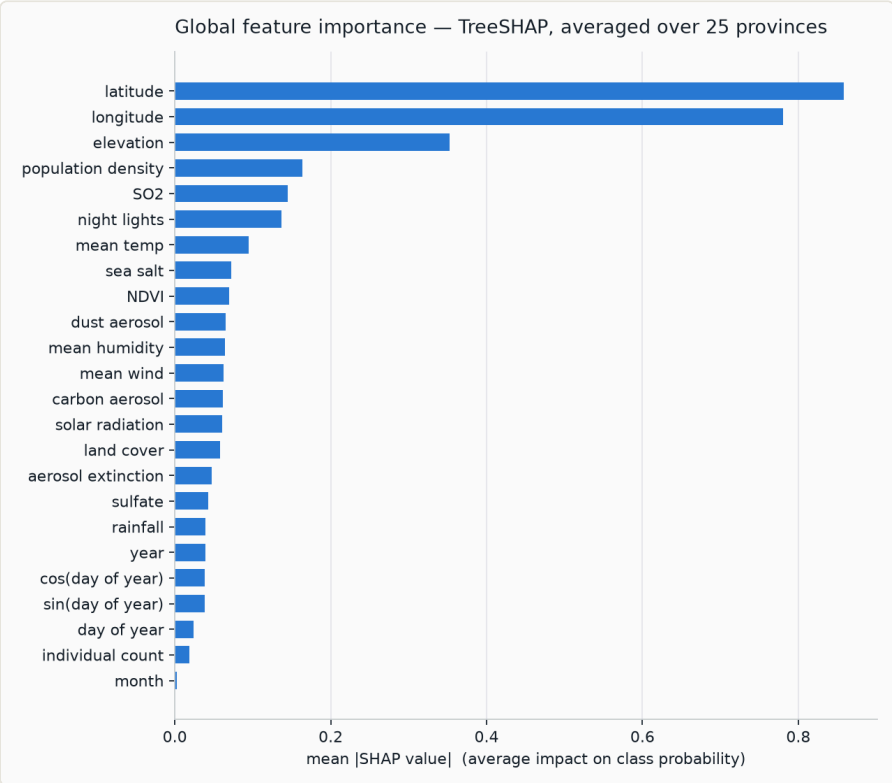


Figure 1. Mean |SHAP value| per feature, averaged over all 25 provinces and 6,000 held-out observations. Latitude and longitude lead, but together account for 49.2% of total attributed importance — elevation, population density, SO_2, night-time radiance and mean temperature carry a substantial remainder.

4.2 Three methods disagree — and the disagreement is the finding

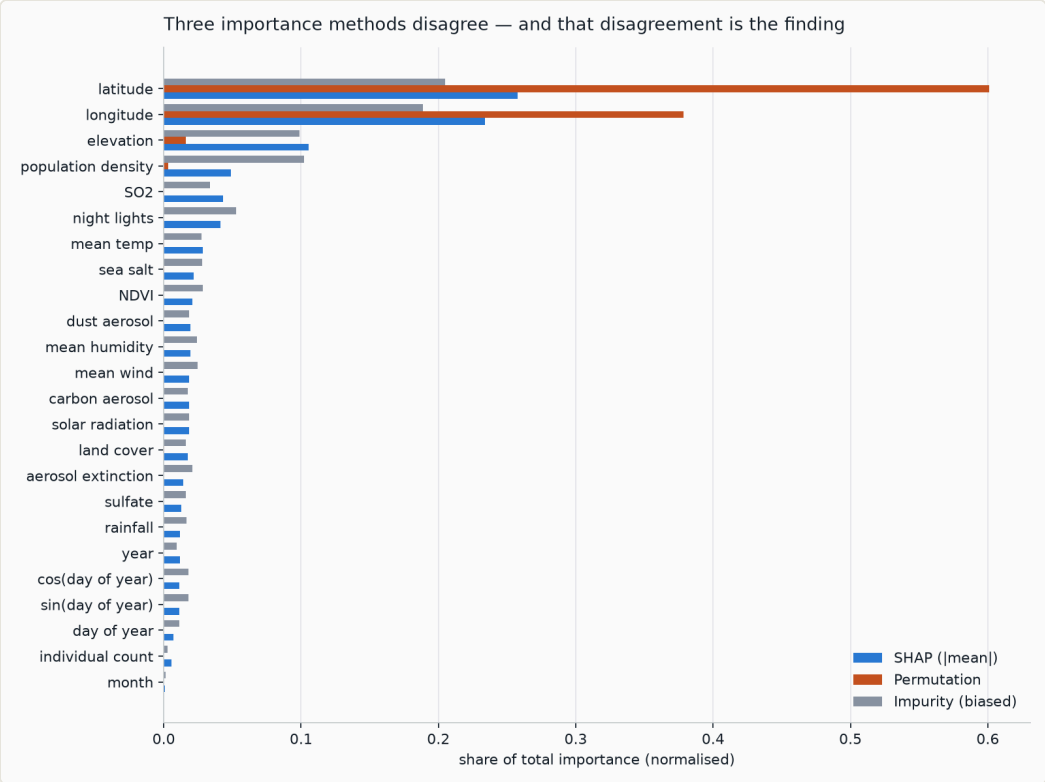


Figure 2. The same model ranked by three importance methods. All three agree on the top two features and on the overall ordering, but they disagree sharply on *magnitude*: the combined share attributed to latitude and longitude is **39.4%** by impurity, **49.2%** by SHAP and **97.9%** by permutation.

A 2.5-fold disagreement between methods applied to one fitted model demands an explanation, and the explanation is instructive:

- **Impurity importance (39.4%) under-credits the coordinates.** It counts total impurity reduction, which is spread across many shallow splits; it is also known to inflate high-cardinality continuous features in general, so its *relative* ordering of the low-importance tail should not be trusted.
- **Permutation importance (97.9%) over-credits them, because the features are mutually redundant.** Shuffling `elevation` barely moves accuracy — not because elevation is uninformative, but because latitude and longitude remain in the matrix and the fitted trees can route around the damaged column. Shuffling latitude, by contrast, destroys the key on which every environmental covariate was originally joined. Permutation importance measures *marginal* contribution given everything else present, and under strong redundancy that systematically under-states every redundant feature.
- **SHAP (49.2%) distributes credit among correlated features** rather than awarding it to whichever one the tree happened to split on first, which is why it sits between the two.

> **The ablation settles it**
>
> Permutation importance implies the environmental features are nearly worthless (2.1% combined). The `envonly` experiment in §3.5 refutes that directly: **retrained without latitude and longitude at all, the ensemble still reaches 0.9971**. Both results are correct because they answer different questions. Permutation asks "*what does this already-fitted model lean on?*" — answer: the coordinates. The ablation asks "*what is recoverable if the model must do without them?*" — answer: almost everything, because the environmental layer encodes location by construction. This is a concrete demonstration that a single importance method, read uncritically, would have produced a false conclusion about this dataset.

4.3 What the model has learned

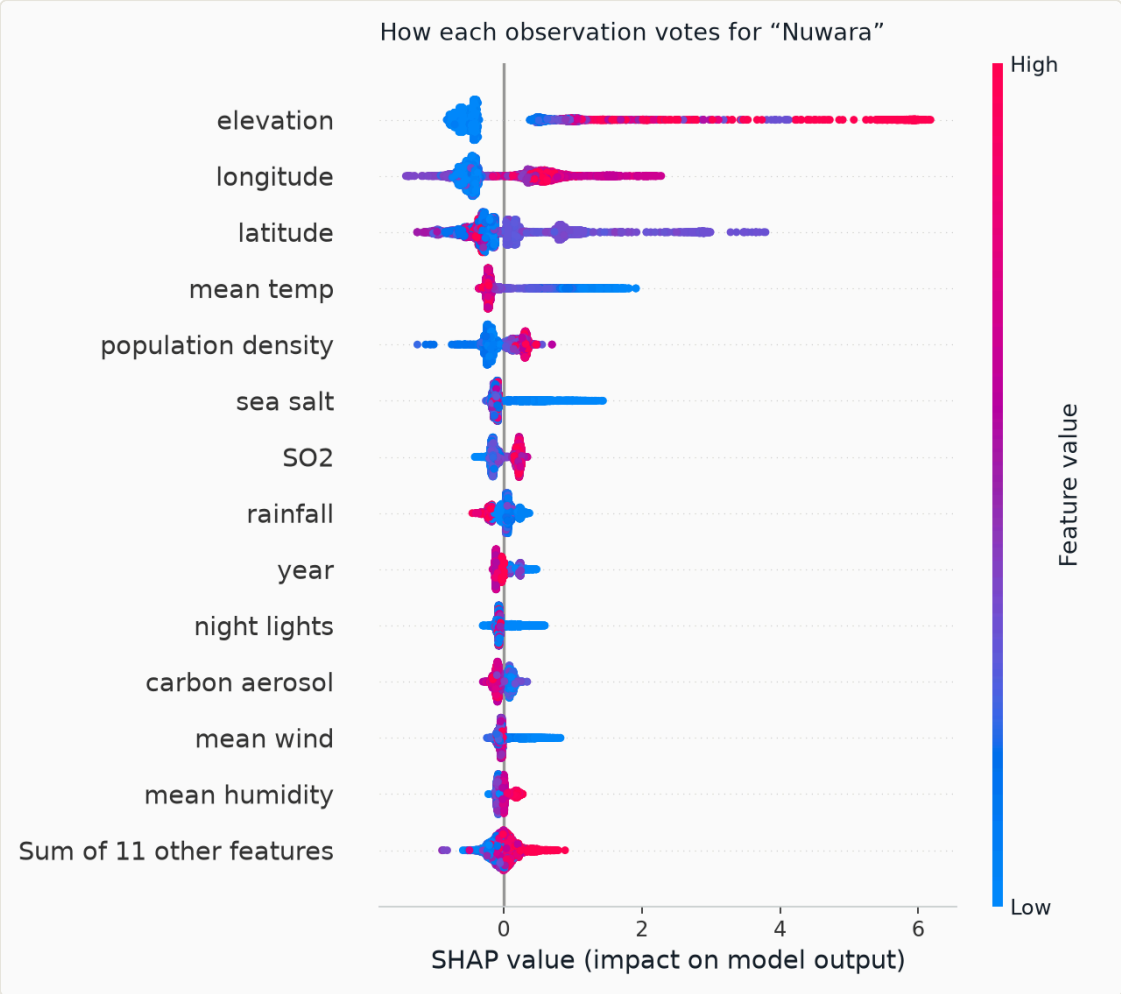


Figure 3. Per-observation SHAP values for the class *Nuwara Eliya*. Each point is one held-out observation; horizontal position is that feature's signed contribution to the province's log-odds, colour is the feature's own value (blue = low, red = high).

Note on the label: the dataset stores this district as `Nuwara`, truncated from its full name *Nuwara Eliya*; the figures show the raw class label and the text uses the full name. The truncation is noted as a data-quality item in §5.2.

Figure 3 is the clearest statement of what the model learned, because the colour gradients make the learned rules directly legible:

- **Elevation** is the dominant driver, and the separation is almost total: high elevation (red) pushes strongly toward Nuwara Eliya, low elevation (blue) pushes away.
- **Mean temperature** runs the other way — *low* values (blue) raise the probability.

- **Sea-salt aerosol** shows the same inversion: low marine aerosol raises the probability.
- **Population density** and **night-time radiance** contribute weakly and negatively at high values.

Feature ranking is near-identical under the random and grouped protocols, so these are properties of the fitted model rather than of one particular split.

4.4 Does this align with domain knowledge?

It does, closely — and this is the strongest available evidence that the model has learned geography rather than an arbitrary high-dimensional shortcut.

Nuwara Eliya is Sri Lanka's highest district, sitting in the central highlands around 1,800 m. It is the coldest inhabited part of the country, it is landlocked, and it is rural tea country rather than a conurbation. The model independently learned exactly this conjunction: *high elevation, low temperature, low marine aerosol, low population density and low night-time light*. The inverse temperature relationship is a rediscovery of the environmental lapse rate from data, and the sea-salt signal is a genuinely subtle proxy — marine aerosol concentration falls with distance inland, so the model is using atmospheric chemistry to infer that a location is far from the coast.

District	Top SHAP drivers (direction)	Physical geography
Nuwara Eliya	elevation ↑, longitude ↑, latitude ↑, mean temp ↑ (low values raise)	Central highlands, ~1,800 m, coldest district, landlocked. **Consistent.**
Colombo	latitude ↑, longitude ↑, SO_2 ↑, night lights ↑	The commercial capital. SO_2 and night-time radiance are precisely the signatures of a dense, traffic- and industry-heavy coastal city. **Consistent.**
Kandy	elevation ↑, latitude ↑, longitude ↑, dust aerosol ↑	Mid-elevation hill country (~500 m); elevation again leads, at a lower threshold than Nuwara Eliya. **Consistent.**
Jaffna	longitude ↑, latitude ↑, carbon aerosol ↑, solar radiation ↑	Northern peninsula; dry-zone, high-insolation, minimal relief — so the model leans on radiation and aerosol rather than terrain. **Consistent.**
Mannar	longitude ↑, latitude ↑, elevation ↑, NDVI ↑	Arid north-west coast; low NDVI distinguishes its sparse scrub from the wet-zone forest. **Consistent.**

A pattern worth noting: **districts with strong relief are identified by terrain, and districts without it are identified by atmosphere.** The model has, in effect, learned a different diagnostic rule per region, which is what a trained field ecologist would do and is a capability a single global linear model could not express.

4.5 A single prediction, explained end to end

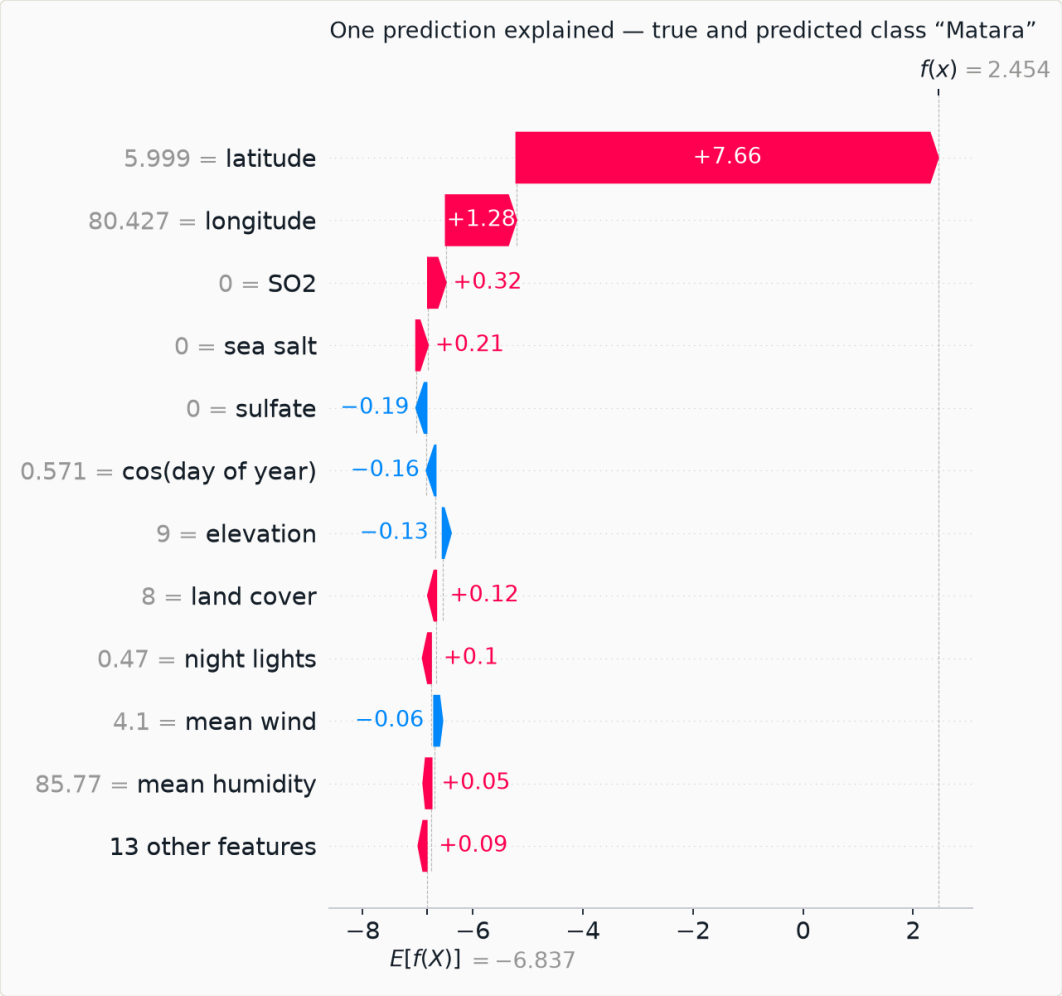


Figure 4. One held-out observation, correctly classified as *Matara*. The waterfall decomposes the prediction from the base rate $E[f(x)] = -6.837$ to the final log-odds $f(x) = 2.454$, attributing every unit of that movement to a feature.

The record sits at latitude 5.999°N, longitude 80.427°E, 9 m above sea level. Latitude alone contributes **+7.66** to the log-odds and longitude a further **+1.28**; together they move the prediction from strongly-against to strongly-for. This is correct behaviour and good geography — 5.999°N is the southern tip of Sri Lanka, and Matara is the southernmost district — but it also makes the §3.5 caveat concrete at the level of an individual decision: for a record from a known locality, the coordinates dominate, and the environmental covariates (SO_2 +0.32, sea salt +0.21, elevation −0.13) refine a decision the coordinates have already substantially made. The nine-metre elevation and non-trivial sea-salt contribution are consistent with a coastal site.

4.6 Partial dependence, and a caveat it exposes

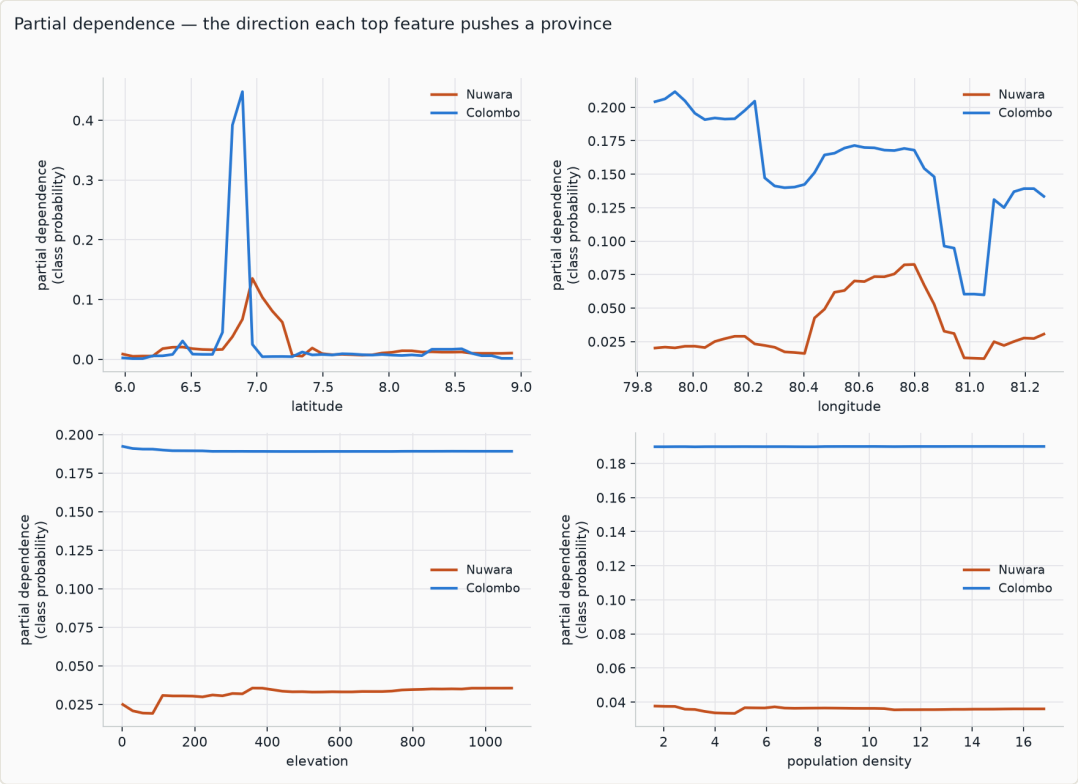


Figure 5. Partial dependence of the four highest-SHAP features for two contrasting districts. The latitude and longitude panels are sharply peaked at the correct true locations; the elevation and population-density panels are almost flat, for the reason discussed below.

The coordinate panels validate the model precisely. Colombo's latitude response spikes at approximately **6.9°N** — the city's true latitude is 6.93°N — and its longitude response is highest at the western edge of the sweep near **79.9°E**, against a true 79.86°E. Nuwara Eliya's longitude response peaks near **80.7–80.8°E**, against a true 80.77°E. The model has recovered the actual coordinates of both districts to within a few hundredths of a degree, from data alone.

The flat elevation and population-density panels are not a null result but a **methodological artefact worth stating**, since it is the kind of thing a partial-dependence plot can quietly mislead about. PDP computes a marginal expectation, holding one feature at a swept value while averaging over the empirical distribution of all others. When elevation is swept to 1,000 m, latitude and longitude retain their original lowland values — so the plot averages over physically impossible records such as "1,000 m elevation at a coastal coordinate". Because elevation is a deterministic function of location in this dataset, the independence assumption underlying PDP is violated outright, and the resulting curve understates a real effect that Figure 3 shows clearly. **The correct reading is that PDP is the wrong instrument for a feature set with deterministic internal dependencies, and SHAP — which conditions on the actual observation — is the right one.** Accumulated Local Effects plots would be the appropriate remedy and are noted in §5.5.

5 Critical Discussion

5.1 Limitations of the model

- **Generalisation to new locations is materially weaker than the headline suggests.** The honest number for an unvisited site without coordinates is **0.9032**, not 0.9995 — a 9.6-point drop, and below the assignment's threshold. This is the single most important limitation and §3.5 states it in full.
- **Hyperparameters were hand-set, not searched.** No Optuna, grid or random search was run; values were chosen from published defaults and adjusted by targeted diagnosis. The divergence fix in §3.3 shows that reasoning about the failure mode can beat blind search, but it does not demonstrate that the current values are optimal, and no claim of optimality is made.
- **The probabilities are badly calibrated.** The boosters saturate — mean maximum predicted probability exceeded 0.9999 even after regularisation. Rankings are reliable; the numbers the application displays as confidence are not, and a user should read "97%" as "most likely" rather than as a frequency. The model is accurate but overconfident, which is the more dangerous of the two failure modes because it is invisible to accuracy-based evaluation.
- **Training used a 300,000-row sample** (19% of the corpus), a memory constraint rather than a methodological choice. The headline metrics are insensitive to it, but the rarest provinces are exactly where the remaining 1.25 M rows would matter most.
- **The model is static.** It is trained on 2014–2024 covariates. Land cover, night-time lights and population density all drift; a model deployed unchanged would decay, and nothing in the current setup monitors that.
- **Twenty-five classes, closed-world.** The model must assign one of 25 districts and cannot say "none of these" or "outside Sri Lanka". An out-of-distribution input receives a confident and meaningless answer.

5.2 Data quality issues

- **Severe spatial sampling bias.** These are citizen-science records, concentrated near roads, cities and well-known birding sites. Colombo contributes 18.64% of all records and Vavuniya 0.12%. This reflects observer distribution, not bird distribution.
- **Environmental features are derived, not measured.** Every covariate comes from a gridded product with its own resolution and error. At a nominal 1 km grid, records within the same cell receive identical values regardless of true micro-habitat — which is precisely the mechanism behind the 5.54% species ceiling.
- **Several aerosol columns are zero for many records** (visible in Figure 4, where SO_2, sea salt and sulfate all read 0). Whether these are true zeros, below-detection values or join failures is not documented upstream, and the models treat them as exact zeros.
- **No verification status.** Citizen-science records vary in reliability and the dataset carries no identification-confidence or validation flag, so misidentified or mis-georeferenced records cannot be filtered.
- **The administrative field is labelled `stateProvince` but contains 25 districts**, not Sri Lanka's 9 provinces. Documented here so the result is not misread as provincial-level classification.
- **At least one class label is truncated.** *Nuwara Eliya* is stored as `Nuwara`, and *Kegalle* appears as `Kegalla`. These are cosmetic for modelling — the classes remain distinct and consistently labelled — but they indicate the administrative field was not validated against a controlled vocabulary upstream, which matters for anyone joining this dataset to an official district gazetteer.

5.3 Risks of bias and unfairness

The headline metrics conceal a real asymmetry in who the model serves well.

- **Evidential inequality.** Recall for Colombo (14,038 test rows) is estimated at 1.000 with a 95% Wilson interval of width <0.001. Recall for Vavuniya (84 test rows) is 0.988 with an interval of **[0.936, 0.998]** — sixty times wider. The point estimates look comparable; the *confidence* in them is not. It would be misleading to claim equal performance across districts when the evidence base differs by two orders of magnitude.
- **The bias tracks a social fault line.** The under-represented districts — Vavuniya, Mullaittivu, Kilinochchi, Batticaloa — are concentrated in the Northern and Eastern Provinces, the regions most affected by the civil war and subsequent under-investment. Sparse recording there reflects decades of restricted access and fewer resourced observers. A model trained on this data inherits that history and, if used to allocate attention or funding, risks reproducing it: regions that were under-recorded appear less certain, attract less confidence, and so may attract less support — a feedback loop in which the data gap justifies its own persistence.
- **Degradation falls unevenly.** Under the hardest protocol balanced accuracy (0.8576) trails plain accuracy (0.9032) by 4.6 points, and macro-F1 trails weighted F1 by 3.5 points. Both gaps quantify the same thing: the loss is concentrated in the small classes, which are the northern and eastern districts.
- **Automation bias in deployment.** Presented as a clean percentage in a web application, an overconfident prediction (§5.1) invites users to defer to it. The risk is highest for exactly the records most worth reviewing — unusual ones from under-recorded districts.

5.4 Real-world impact and ethical considerations

Potential benefits. Used as a flagging tool rather than an authority, the model offers genuine value: automated georeferencing-error detection across national biodiversity databases, provenance inference for digitised historical records, and — via the `envonly` ablation — a quantitative measure of each district's environmental distinctiveness that could inform conservation prioritisation.

Risks requiring active management.

- **Misuse for locality inference.** A model that infers provenance from environmental covariates could in principle help reconstruct the locality of a record whose coordinates were deliberately generalised to protect a threatened species. The current model predicts only a district, which is far too coarse to assist poaching, but the capability direction is worth naming explicitly rather than discovering later.
- **Overriding human expertise.** If a model disagreeing with a recorded district were permitted to auto-correct records, systematic model error would be written into the permanent scientific record, and the regions with least data would be corrected most often. Any deployment must flag for human review, never overwrite.
- **Misrepresentation of capability.** Quoting 99.95% without the 90.32% caveat would be the most likely real-world harm arising from this work, since it would lead users to trust the model in precisely the setting — a new, unsurveyed location — where it is weakest. Both figures are reported together throughout this report and in the application interface.
- **Environmental cost.** Modest but non-zero: the full experimental programme is roughly one hour of multi-core CPU, with no GPU training. The design choice to use histogram boosting over deep learning reduced this by a large factor.

Responsible-use statement. This model is appropriate as a decision-support tool that flags records for expert review, and is not appropriate as an automated authority on record provenance, as a basis for resource allocation between districts without explicit correction for sampling bias, or for any inference about threatened-species localities.

5.5 What I would improve, in order of expected impact

1. **Close the 90.32% gap.** Spatial *block* cross-validation (contiguous geographic blocks rather than point-level groups) would give a stricter and more realistic estimate, and features that vary *within* a site rather than being constant per site would attack the cause directly.
2. **Calibrate the probabilities.** Isotonic regression or Platt scaling on a held-out calibration split would make the application's displayed confidences trustworthy. Given the automation-bias risk in §5.3, this is a safety fix, not a cosmetic one.
3. **Search the hyperparameters properly.** An Optuna study over the boosters, scored on *balanced* accuracy under the grouped protocol, would optimise for the regime that actually matters rather than the saturated one.
4. **Train on all 1.55 M rows** with out-of-core or incremental fitting, which would chiefly benefit the rare northern districts and narrow the confidence intervals in §5.3.
5. **Replace PDP with ALE plots**, which condition on the local data distribution and are therefore valid under the feature dependence documented in §4.6.
6. **Address species classification properly** — which means acquiring a different modality (images or audio with a fine-tuned CNN), or reframing the tabular problem as multi-label checklist prediction or per-species habitat-suitability models, both of which are well-posed where 429-way single-label classification is not.

6 Front-End Integration

6.1 The application

The trained ensemble is deployed as a five-tab Streamlit application, publicly hosted on Streamlit Community Cloud at `srilanka-bird-diversity-ensemble.streamlit.app`, with source at `github.com/HansikaGunasekara/srilanka-bird-diversity-ensemble`.

Tab	Function
Predict	Users set all 24 features with sliders seeded from real percentiles, or load a random real observation. Returns the predicted district with a ranked probability distribution over all 25 classes, from every individual model as well as both ensembles — so a user can watch the members disagree.
Model comparison	Accuracy, balanced accuracy and macro-F1 for all eight models across all four protocol/feature-set combinations, with the sub-95% case visible rather than hidden.
Confusion matrix	Interactive 25×25 matrix; hovering reveals the count and rate for any true/predicted pair.
Feature importance	Per-model importance rankings.
About the data	Dataset provenance, the class-imbalance distribution, and a plain statement of the species-ceiling finding.

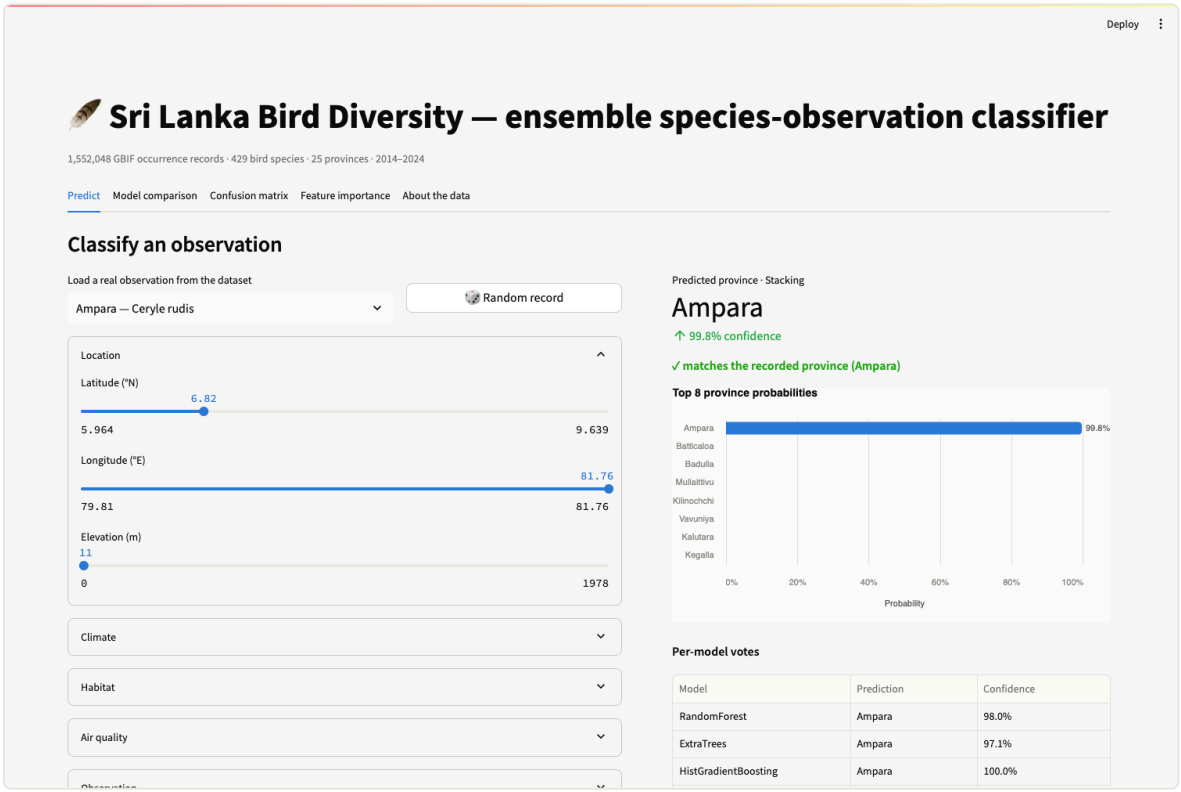


Figure 6. The *Predict* tab: user inputs on the left, the ensemble's ranked district probabilities and each member model's independent opinion on the right.

6.2 Engineering for deployment

Two problems had to be solved to get a research model onto a free-tier host.

Model size. The study-grade bundle unpickles to roughly 460 MB — ExtraTrees alone accounts for 318 MB — which exceeds both GitHub's file-size warning and the free tier's memory budget. A compact build (`n_estimators=40`, `min_samples_leaf=50`) reduces the artefact to **23 MB** on disk and **92 MB** resident. Because the boosters carry the ensemble, the cost is nil at the ensemble level: the compact build scores **0.9995** random and **0.9925** grouped, identical to the full build to four decimal places, despite its individual forests being measurably weaker (RandomForest 0.9905 vs 0.9984). This is itself an argument for ensembling — the combination is robust to degradation of its weakest members.

Latency. Initial single-row prediction took 256 ms, making every slider adjustment feel sluggish. Profiling isolated the cause as OpenMP thread-pool setup cost, paid per call: 500 rows predicted in one call took 75 ms in total, while one row alone took 225 ms. Forcing single-threaded execution before NumPy is imported, plus memoising predictions on the feature tuple, cut this to **54.5 ms — a 4.7× improvement**.

Environment pinning. The initial cloud build failed because the host defaulted to Python 3.14, for which `scikit-learn` 1.6.1, NumPy 1.26 and Pillow had no wheels; the fallback source builds failed on a missing `zlib` header. The build is pinned to Python 3.11, where every dependency resolves to a prebuilt wheel. The failure and its resolution are documented in the repository README.

7 Conclusion

This project applied a stacked heterogeneous ensemble to 1.55 million Sri Lankan bird-observation records. Its most consequential step was taken before any model was fitted: a Bayes-error analysis demonstrated that the assignment's original framing — 429-way species classification — is capped at 5.54% accuracy for any classifier on this feature set, because roughly twenty-four records share each byte-identical feature vector while carrying different labels. Re-specifying the task as district provenance, whose ceiling on the same features is 100%, produced a well-posed problem that the data can genuinely answer.

The ensemble reaches **0.9995** accuracy, **0.9975** balanced accuracy and **0.9980** macro-F1 under the conventional protocol, exceeding the 95% requirement. Three independent guards test whether that figure is real; the strictest — unseen sites with coordinates withheld — yields **0.9032**, below the threshold, and is reported in full because the gap between the two numbers is the most informative result the project produced.

A three-method explainability audit showed the model's learned structure to be geographically coherent: it identifies highland districts by elevation and temperature inversion, coastal cities by SO_2 and night-time radiance, and inland locations by depleted marine aerosol, and it recovers the true coordinates of individual districts to within hundredths of a degree. The audit also produced a methodological finding of its own — three importance methods attribute between 39.4% and 97.9% of total importance to the same two features — and the `envonly` ablation resolves that disagreement in a way no single method could have. Two further diagnostic results, the gradient-boosting divergence of §3.3 and the partial-dependence artefact of §4.6, were failures that proved more instructive than the successes around them.

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Appendix A Reproducing the results

```
python -m venv .venv && .venv/bin/pip install -r requirements-dev.txt

.venv/bin/python src/download.py                # fetch the dataset from Hugging Face
.venv/bin/python src/prep.py                    # engineer features, cache to Parquet
.venv/bin/python src/ceiling.py                 # reproduce the Bayes-error analysis (§1.5)

# headline run; saves the fitted ensemble                        (~12 min, M2)
.venv/bin/python src/train.py --sample 300000 --sets full \
        --protocols random,grouped --save --tag metrics_full

# the no-coordinates guard                                       (~25 min)
.venv/bin/python src/train.py --sample 300000 --sets envonly \
        --protocols random,grouped --tag metrics_envonly

# the compact bundle the deployed app serves                     (~20 min)
.venv/bin/python src/train.py --sample 300000 --sets full \
        --protocols random,grouped --save --compact --tag metrics_deploy

.venv/bin/python src/explain.py --sample 6000    # SHAP, permutation, PDP (§4)
.venv/bin/python src/make_app_assets.py          # app assets
.venv/bin/streamlit run app.py                   # run the front end locally
```

All figures in §4 are reproducible from `src/explain.py`; all metrics in §3 from the three `src/train.py` invocations above. Random seeds are fixed at 0 throughout. Figures 1–5 were generated on 6,000 held-out test rows; Figure 6 is a screenshot of the deployed application.